Matrix product state simulator aborts with wrong error message

https://github.com/Qiskit/qiskit-aer/issues/1920

ROW 118

Matrix product state simulator aborts with wrong error message #1920

New issue

Closed

smiserman opened on Aug 29, 2023 · edited by smiserman

Edits ···

Informations

- **Qiskit Aer version: 0.12.1:**
- **Python version: 3.10.9:**
- **Operating system: Win11, 16 GB RAM:**

What is the current behavior?

The `AerSimulator(method='matrix_product_state')` aborts with an error message if the number of qubits of the circuit is greater than the maximum memory / 32 MB, regardless of the size of a generated MPS. With 16 GB RAM the number of qubits is limited to 504, which is in any case not necessary with a moderate bond dimension.

Steps to reproduce the problem

```
num_qubits = 505
circ = QuantumCircuit(num_qubits)
circ.h(0)
for i in range (0, num_qubits-1):
    circ.cx(i, i+1)
circ.measure_all()

simulator = AerSimulator(method='matrix_product_state')
res = simulator.run(circ).result()
```

gives

```
Simulation failed and returned the following error message:
ERROR:  [Experiment 0] Insufficient memory to run circuit circuit-154 using the matrix_product_state
simulator. Required memory: 16160M, max memory: 16133M
```

What is the expected behavior?

The simulation of the circuit above works fine in 1-2 sec for 504 qubits. So it should not abort with 505 qubits. Furthermore the required memory for the MPS is only about 0.3 MB and not $505 \cdot 32 = 16160$ MB!

Suggested solutions

You should keep track of the size of the generated MPS, which is bounded above by approximately $N(32d^2 + 8d)$ bytes, where N is the number of qubits and d is the maximum bond dimension. There is some overhead for list, tuple, and Ndarray structures that can be neglected for larger d. In my own simulation, run in Numpy, I get $N(32d^2 + 8d + 440)$ and which, interestingly, is several seconds faster than the qiskit simulation for larger d. So one should be able to comfortably simulate 1000 qubit circuits with d = 64 (correspond to 6 entanglement blocks in a VQE ansatz) and even more so with d=2 on a 16 GB RAM device.

Another remark

You should adapt your matrix product state simulation method tutorial in qiskit: *We can handle more qubits than this, but execution may take a few minutes. Try running a similar circuit with 500 qubits! Or maybe even 1000 (you can get a cup of coffee while waiting).*

This is definitly not true! With constant bond dimension the matrix product state algorithm scales linearly with the number of qubits and the number of shots (as long shots $\gg d$). So if it takes about 0.1 seconds for 50 qubits, it takes about one second for 500 qubits and two seconds for 1000 qubits. That's not a few minutes and certainly not time to drink a coffee! And this is not only theoretical, I checked the linearity practically on my as well as on Aer Simulation.

smiserman added bug on Aug 29, 2023

smiserman changed the title ~~Matrix product Simulator aborts with wrong error messa~~ Matrix product Simulator aborts with wrong error message on Aug 29, 2023

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Assignees

doichanj

Labels

bug

Type

No type

Projects

No projects

Milestone

Aer 0.13.0

Closed on Jan 10, 2024, 82% complete

Relationships

None yet

Development

No branches or pull requests

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Participants

Goodnotes

smiserman added bug on Aug 29, 2023

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doichanj on Aug 30, 2023 — Collaborator

qiskit-aer/src/simulators/matrix_product_state/matrix_product_state.hpp
Lines 321 to 331 in ada56a3

```
321     size_t State::required_memory_mb(uint_t num_qubits,
322                                      const std::vector<Operations::Op> &ops) const {
323       // for each qubit we have a tensor structure.
324       // Initially, each tensor contains 2 matrices with a single complex double
325       // Depending on the number of 2-qubit gates,
326       // these matrices may double their size
327       // for now - compute only initial size
328       // later - FIXME
329       size_t mem_mb = 16 * 2 * num_qubits;
330       return mem_mb;
331     }
```

I think this function returns required memory size in byte not in MB, but by dividing this by 1000000 it returns 0.
Comment in this code says FIXME, so this should be fixed

doichanj added this to the Aer 0.13.0 milestone on Aug 30, 2023

smiserman on Aug 30, 2023 — Author

Ok, but if you just divide this by 1000000, this is a very optimistic assumption without any information ;-). And in my opinion, the size of the MPS is not determined by the number of 2-qubit gates, but by the number of entanglement blocks (you can add any number of CNOT gates for the above EPR state without changing the bond dimension from 2) . However, you don't know this in advance, so you have to keep the maximum bond dimension tracked while calculating the MPS from the circuit, as I suggested.

doichanj self-assigned this on Sep 4, 2023

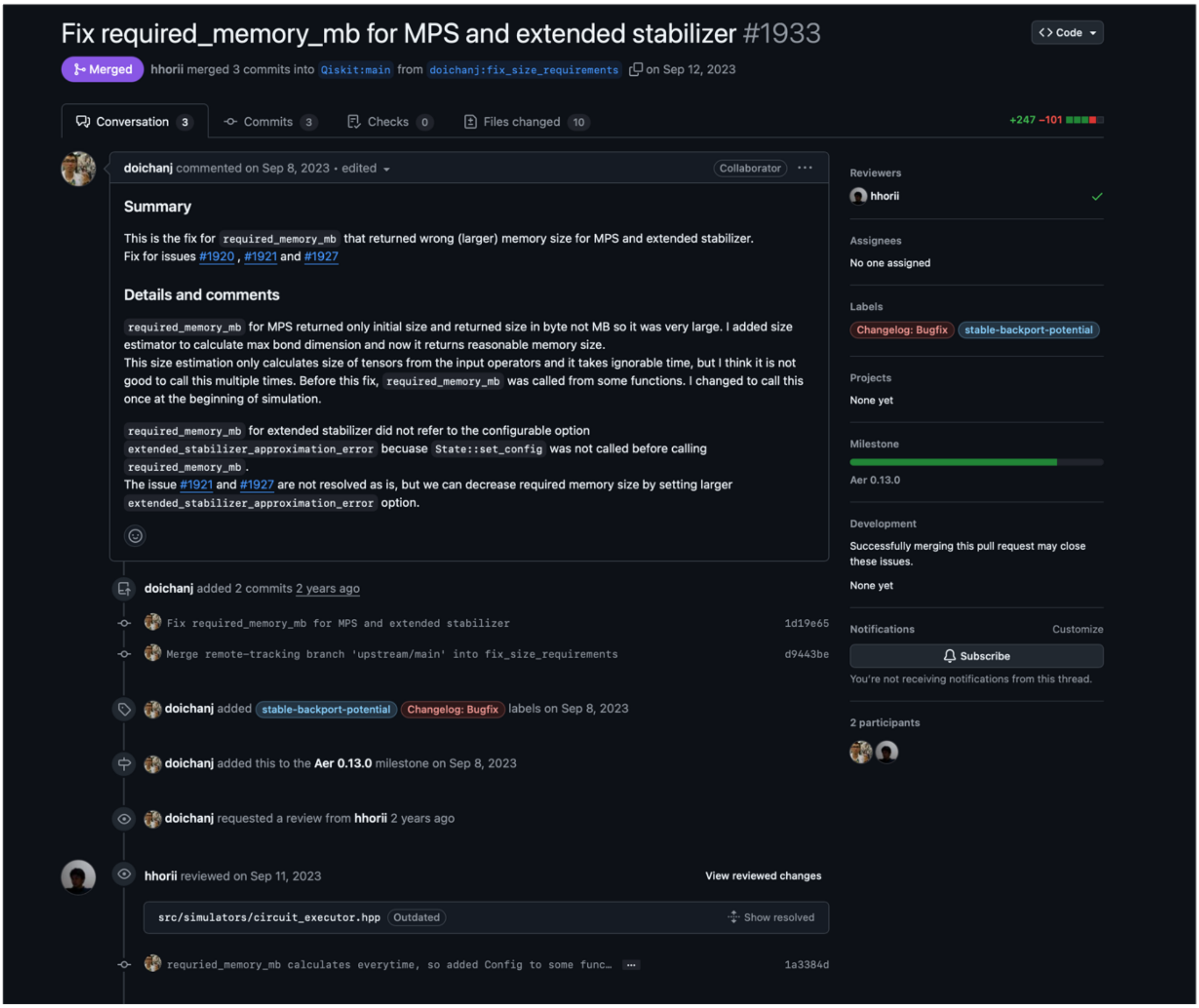
doichanj mentioned this on Sep 8, 2023

Fix required_memory_mb for MPS and extended stabilizer #1933

doichanj on Sep 15, 2023 — Collaborator

I think this should be fixed by PR #1933

doichanj closed this as completed on Sep 15, 2023



Fix required_memory_mb for MPS and extended stabilizer #1933

Code

Merged hhorii merged 3 commits into `Qiskit:main` from `doichanj:fix_size_requirements` on Sep 12, 2023

Conversation 3 | Commits 3 | Checks 0 | Files changed 10 | +247 −101

doichanj commented on Sep 8, 2023 • edited — Collaborator

Summary

This is the fix for `required_memory_mb` that returned wrong (larger) memory size for MPS and extended stabilizer.
Fix for issues #1920 , #1921 and #1927

Details and comments

`required_memory_mb` for MPS returned only initial size and returned size in byte not MB so it was very large. I added size estimator to calculate max bond dimension and now it returns reasonable memory size.
This size estimation only calculates size of tensors from the input operators and it takes ignorable time, but I think it is not good to call this multiple times. Before this fix, `required_memory_mb` was called from some functions. I changed to call this once at the beginning of simulation.

`required_memory_mb` for extended stabilizer did not refer to the configurable option `extended_stabilizer_approximation_error` becuase `State::set_config` was not called before calling `required_memory_mb` .
The issue #1921 and #1927 are not resolved as is, but we can decrease required memory size by setting larger `extended_stabilizer_approximation_error` option.

doichanj added 2 commits 2 years ago

`Fix required_memory_mb for MPS and extended stabilizer` 1d19e65

`Merge remote-tracking branch 'upstream/main' into fix_size_requirements` d9443be

doichanj added stable-backport-potential Changelog: Bugfix labels on Sep 8, 2023

doichanj added this to the **Aer 0.13.0** milestone on Sep 8, 2023

doichanj requested a review from **hhorii** 2 years ago

hhorii reviewed on Sep 11, 2023 — View reviewed changes

`src/simulators/circuit_executor.hpp` Outdated — Show resolved

`requried_memory_mb calculates everytime, so added Config to some func…` 1a3384d

Reviewers: hhorii

Assignees: No one assigned

Labels: Changelog: Bugfix, stable-backport-potential

Projects: None yet

Milestone: Aer 0.13.0

Development: Successfully merging this pull request may close these issues. None yet

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2 participants